



Validity and repeatability of a new inertial measurement unit system for gait analysis on kinematic parameters: Comparison with an optoelectronic system

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ABSTRACT

Inertial measurement unit (IMU) systems are offering promising alternatives to optoelectronic systems for measuring human movement. The objective of this study was to assess the validity of a new inertial measurement system, iSen STT-IWS sensors, with reference to the optoelectronic motion capture system OptiTrack. 22 young participants performed three minutes of treadmill walking at different speeds. Hip, knee and ankle joint angles in the sagittal plane were compared between the two systems. Accuracy and repeatability of iSen IMU measures was evaluated with the Root Mean Square Deviation (RMSD) and Lin's Concordance Correlation (LCC). A good to tolerable iSen measures' accuracy was found (RMSD < 10°, LCC > 0.75) and reliability decreased with walking speed. Results indicated excellent test-retest repeatability of the inertial system (Intraclass correlation (ICC) > 0.75). Sagittal kinematic gait estimation using the iSen IMU system can be reliably used for clinical purposes on older adults.

1. Introduction

Quantitative Gait Analysis (QGA) is a method allowing to assess the quality of an individual's gait by providing objective measurement characterizing the walking pattern. This technic opens the field of possibilities in terms of clinical evaluation of gait disorders as it is more accurate than the visual observation by the therapist usually used in some clinical tests such as Berg Balance Scale or six-minute walk test. QGA can be performed complementary to a clinical observations where spatio-temporal parameters are calculated such as walking speed, stride length and cadence. This analysis is usually performed on vulnerable patients to determine functional limitations or pathologies – orthopaedic, neurologic or traumatic – affecting gait parameters [1] and facilitate treatment decision making. Studying human movement and understanding the biomechanical aspects are essential to improve rehabilitation treatments [2].

Ever since their development in the 1990s, motion capture

technologies have been continually evolving. They are based on increasingly varied approaches providing more accurate and realistic results. Optoelectronic motion capture systems (MOCAP) are used to quantify movement kinematics using reflective sensors placed on the body and multiple infra-red cameras tracking the three-dimensional trajectory of the markers. They are highly accurate and often considered as laboratory gold standards in motion analysis as they are the most commonly used technique to study lower limb movement [3]. However, these MOCAP exhibit limitations [4]. Indeed, they required technical or clinical staff trained to use such an equipment, they are also time-consuming, very expensive and restrictive in the capture volume by the cameras making them of little practical use for day to day evaluation in clinics. Furthermore, the accuracy in the obtained joint kinematics directly relates to a correct placement of markers and soft tissue artifacts [3–5]. There is, thus, a need to overcome these limitations and to simplify and making more accessible the monitoring of human movements.

Recent reviews have highlighted the growing interest in inertial

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measurement units (IMUs) as an alternative to MOCAP [6]. IMUs are sensors composed of gyroscope, an accelerometer, and a magnetometer and are able to estimate motion through the location and the orientation of the sensors. Recently, a full inertial motion capture system (iSen, STT Systems Inc., San Sebastian, Spain) was developed giving the possibility to choose the appropriate set up (composed to 2 to 16 IMUs) based on the clinical application. It offers a real-time feed-back, a short setup time, and a relatively low cost compared to MOCAP system. Previous studies have previously validated similar systems [4,7,8] while no study have validated this specific ISEN system and quantify the possible bias of IMUs and kinematics errors. Although IMUs are seen as promising solutions for motion capture, they are more prone to errors that could affect kinematics detection. Many studies have reported the effect of sensors drift caused by the accumulation of gyroscope errors over time [9,10,11] while other studies indicated offset problems [12]. It is thus important to assess the reliability of IMUs kinematic measurements with reference to a “Gold standard” system before a clinical use.

In this context, the aim of this study was to compare two commercially available motion capture systems. iSen IMU sensors were evaluated whereas OptiTrack optoelectronic motion capture system (NaturalPoint, Corvallis, OR USA) was used as gold standard. The first hypothesis was that the ISEN inertial motion capture system can measure 3D lower limb kinematics (hip, knee, ankle) with an acceptable level of error under 3° in the sagittal plane. The second hypothesis was that ISEN measures are highly reliable with ICC higher than 0.7.

2. Materials and methods

2.1. Participants

Twenty-two healthy participants (13 women, 9 men, mean age 26 years $\pm$ 5.4; mean weight 69.6 kg $\pm$ 14.8, mean height 171.8 cm $\pm$ 8.2, mean BMI 23.4 kg/m² $\pm$ 3.4) were recruited. Values are mean $\pm$ Standard Deviation (SD). Exclusion criteria was musculoskeletal injuries.

2.2. Experimental protocol

2.2.1. Gait protocol

Static calibration was performed to calibrate both the IMU and MOCAP systems. The static calibration estimates the sensors and segments orientation and is used to quantify the initial offsets between the two systems. After five minutes of familiarization with treadmill walking [13], participants walked at three different speeds (0.8, 1.2 and 1.6 m/s) for three minutes at each speed randomly chosen. Speed of 0.8 m/s was chosen first as it represents a valuable indicator of the falling risk among older adults [14]. For a test/retest of the inertial system, participants repeated the same testing procedure a second time, 2 weeks later. The 2 weeks gave sufficient time to recover and to reduce any custom to the treadmill walking.

2.2.2. Data recording and processing

Joint kinematics was captured using eleven IMU sensors (5.6 cm $\times$ 3.8 cm $\times$ 1.8 cm) positioned on the subject according to the «Lower body (with heels) plus spine» model of the iSen software. They were connected to the computer via a wireless Wi-Fi connection. IMU were fixed on the subject with adhesive scratch bands on the feet (rearfoot and forefoot), shanks, thighs, pelvis, chest and back (Fig. 1). Subjects were asked to remove their shoes before beginning the test, so that the IMU could be placed directly on the feet. iSen data were sampled at 100 Hz. Acquisitions were recorded simultaneously at 100 Hz by nine infrared cameras using OptiTrack Motive 2.2.0 (NaturalPoint, Inc., Oregon, USA). Ground reaction forces were collected at 1000 Hz with two force plates under the treadmill. Heel-strike recorded with MOCAP force plates was used for cycle segmentation on knee, hip and ankle joint at each gait speed. Thirty-two passive reflective markers were placed

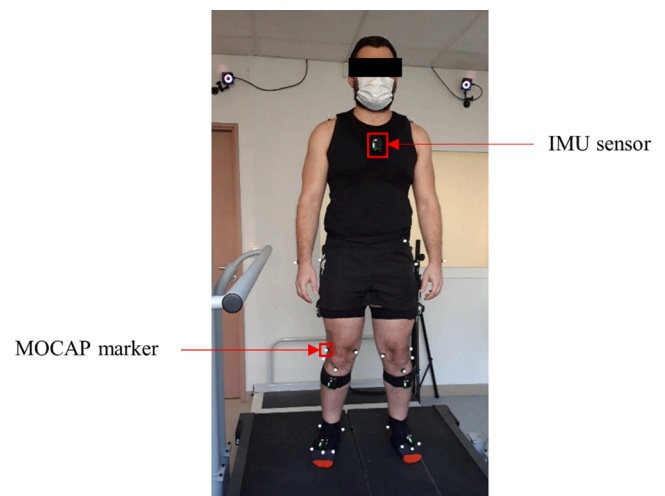


Fig. 1. Experimental setup including 11 IMU, 32 reflective markers, 9 infra-red cameras and 2 force plates under the treadmill.

with double-sided hypoallergenic adhesive tape on the anatomical landmarks of the subject, according to Rizzoli Lower Body markerset protocol in Motive (NaturalPoint Product Documentation, [15]). Three more markers were placed on shoulders, elbows, and wrists (Fig. 1).

Data were recorded simultaneously with the inertial and the MOCAP system and to synchronize the two signals, a cross-correlation between the norms of the angular velocities of the knee from the two systems was applied using Matlab (Version R2018b. Natick, Massachusetts: The MathWorks Inc). Data obtained with the optoelectronic system were exported as C3D files and kinematics calculations were conducted using MATLAB. A 3D labelled model was created, using static and dynamic recordings as well as anthropometrics measurements of the subject (height and weight). Concerning ISEN system, kinematic calculation was made directly on the software and exported as files. The entire acquisition was divided into normalized gait cycle and for each gait cycle, their minimum, maximum and range of motion were taken. From these values, the averaged min, max and ROM was calculated for every joint angles in the sagittal plane.

2.3. Data analysis and statistical calculation

To assess the validity of IMU system, Root Mean Square Deviation (RMSD) was computed on kinematic walking cycle. Lin's Concordance Correlation (LCC) with 95% CIs was calculated to determine the strength of the agreement. Normality of the data was checked using Shapiro-Wilk test. RMSD and LCC calculations were conducted via MATLAB. Bland-Altman analysis was performed with XLSTAT (XLSTAT version 2021.2.2) to analyze bias and limits of agreement. For test-retest reliability, a comparison of gait parameters was made between first and second trials using interclass correlation coefficient (ICC) of type (2,1). SPSS 23.0 statistical software was used for the calculation of ICC. Two-Way Random Model was considered. A two-way repeated-measure analysis of variance (ANOVA) was performed to compare joint angles. Statistically significant p-value was set at 0.05. Clinical significance was set for differences $> 5^\circ$ [12,16].

3. Results

3.1. Validity

The differences in joint angles using IMU and MOCAP systems were not significant except for the range of motion (ROM) of hip-joint angle at 1.2 m/s ($F(1,40) = 15.91$, $p < 0.01$) and 1.6 m/s ($F(1,40) = 6.60$, $p = 0.01$) and maximum ankle-joint angle at 1.2 m/s ($F(1,38) = 2.51$, $p =$

0.01) and 1.6 m/s ($F(1,38) = 6.60, p = 0.00$) (Table 1). RMSD of all joint angles for the points of interests (min, max, ROM) in the sagittal plane are presented in Table 1. RMSD values ranged between 3.3 and 8.4 for all joints and all speeds. RMSD averaged for the three gait speeds were smaller for knee joint (4.3°) than for hip (7.3°) and ankle (7.3°) joints. When analyzing the entire waveform for ankle, knee and hip joint, RMSD was also smaller for knee joint (3.3°) than for hip (7.3°) and ankle (5.6°) joints (Fig. 3). LCC was higher than 0.75 for all joints except for the ankle (0.72 at 0.8 m/s, 0.60 at 1.2 m/s and 0.56 at 1.6 m/s). Strongest agreement was obtained for knee at low speeds ($LCC > 0.75$). Bland-Altman plots in Fig. 2 provide a visual representation of agreement between iSen and OptiTrack. Data appeared within the 95 % limits of agreement. Bias was constant for knee and hip regardless of the speed. IMU system underestimated knee flexion by 4° and overestimated hip flexion by 2° . However, an increase of bias on ankle measurements was observed with gait speed. Smallest bias was observed on ankle dorsiflexion at 0.8 m/s (Max bias = -4.15° , Min bias = -0.26° , ROM bias = -3.89°). Correction of the offsets on knee and ankle angles waveforms, according to Bland-Altman biases, decreased RMSD ($RMSD < 3^\circ$) (Table 1 and Fig. 3). Table 2..

3.2. Gait speed effect

Gait speed effect on kinematics was also analyzed and joint angles waveforms are presented in Fig. 4. Results showed an increase in RMSD and a decrease in LCC. Bias on ankle measurements increased with gait speed (bias on maximum values of -4.15° , -8.35° , -11.73° respectively at 0.8, 1.2 and 1.6 m/s). LCC decreased with gait speed (0.88 at 0.8 m/s, 0.86 at 1.2 m/s and 0.85 at 1.6 m/s). The effect of gait speed was only significant on hip extension ($F(2,122) = 43.759; p < 0.001$) and ankle plantarflexion ($F(2,119) = 12.017; p < 0.001$) captured by the iSen. Same results were obtained for OptiTrack system.

3.3. Repeatability

Test-retest reliability was verified with ICC. Results are presented in Table 3. ICC was higher than 0.75 for all joints and speeds apart from hip extension at 1.6 m/s ($ICC = 0.68$). Average ICC for all speeds were higher for the knee ($ICC = 0.93$) than in hip ($ICC = 0.89$) and ankle joints ($ICC = 0.80$).

3.4. Drift

The comparison of knee, hip and ankle angles between the first and last 30 s of each acquisition showed small relative errors (Table 4). Relative error was calculated as the difference of IMU and MOCAP between the last 30 s and the first 30 s (relative error = $\Delta_{[150:180s]} - \Delta_{[0:30s]}$ with $\Delta = MOCAP - IMU$). Highest relative error was obtained on hip flexion (1.1°). No significant difference was highlighted. Mean gait cycles on these two time periods were similar ($RMSD < 2^\circ$).

- Joint angle difference was calculated on the minimum (min), maximum (max) and range of motion (ROM) values between the first and last 30 s of the acquisition: max/min/rom angle difference = max/min/rom $_{[150:180s]} - \text{max/min/rom}_{[0:30s]}$
- RMSD were calculated comparing IMU mean gait cycle between first and last 30 s.
- max/min/rom relative error = $\Delta_{[150:180s]} - \Delta_{[0:30s]}$ with $\Delta = \text{max/min/rom (MOCAP)} - \text{max/min/rom (IMU)}$ calculated the difference between IMU and MOCAP between first and last 30 s.
- RMSD relative error = $RMSD_{[150:180s]} - RMSD_{[0:30s]}$

4. Discussion

The objective of the study was to validate iSen IMU kinematics with reference to the optical motion capture system (OptiTrack). The accuracy of iSen IMU was evaluated by comparing data obtained with both systems during treadmill walking. Results indicate that iSen has an acceptable and tolerable accuracy in the evaluation of lower-limb kinematics during gait in the sagittal plane. IMU measurements showed excellent repeatability during test-retest experiments.

No significant difference was found between discrete variables (max, min and range of motion) of knee and ankle joint, regardless of the speed. IMU system captured well knee kinematic changes induced by walking speed. However, hip range of motion and ankle dorsiflexion were significantly different at high speed (1.2 and 1.6 m/s) showing a lower accuracy for those parameters. Park and Yoon et al [17] also reported significant differences on ankle dorsiflexion and plantarflexion ($p = 0.01$) at a similar speed (1.34 m/s). This difference impacted hip and ankle RMSD values ($RMSD = 7.3^\circ$) as they are higher than knee RMSD ($RMSD = 4.3^\circ$). However, acceptable accuracy between iSen IMU and OptiTrack MOCAP was found for knee angle ($RMSD = 3.3^\circ$ at 0.8 m/s, $RMSD = 3.9^\circ$ at 1.2 m/s) based on RMSD characterization [16 7].

Table 1

Validity of the IMU against MOCAP. Minimum (min), maximum (max) and range of motion (ROM) values for knee flexion/extension, hip flexion/extension and ankle dorsiflexion/plantarflexion recorded by iSen and OptiTrack at different speeds are presented. Root mean square deviation (RMSD), Lin's concordance correlation coefficient (LCC) and p-value are also presented. NS stands for "not significative".

Joint (Sagittal plane)	Speed [m/s]	Max [°]			Min [°]			Rom [°]			RMSD [°] (±SD)	LCC (95% CI)	Corrected RMSD [°]
		IMU	MOCAP	p-value	IMU	MOCAP	p-value	IMU	MOCAP	p-value			
Knee	0.8	55.72	60.55	NS	9.28	9.74	NS	46.44	50.81	NS	3.3 (±1.86)	0.96 (0.96–0.97)	1.6
	1.2	59.77	64.36	NS	8.42	8.32	NS	51.35	56.04	NS	3.9 (±2.54)	0.96 (0.95–0.97)	2.4
	1.6	59.62	63.93	NS	7.25	8.58	NS	52.38	55.34	NS	5.4 (±3.36)	0.88 (0.86–0.90)	2.5
Hip	0.8	25.08	23.15	NS	−3.57	−10.68	NS	28.65	33.83	NS	7.3 (±5.07)	0.80 (0.75–0.83)	–
	1.2	27.36	26.05	NS	−8.24	−15.51	NS	35.60	41.55	< 0.001	7.8 (±4.72)	0.78 (0.74–0.82)	–
	1.6	21.94	29.68	NS	−13.84	−16.94	NS	41.79	46.63	< 0.001	6.8 (±4.31)	0.84 (0.81–0.88)	–
Ankle	0.8	12.99	8.85	NS	−15.46	−15.72	NS	28.45	24.56	NS	5.3 (±3.61)	0.72 (0.67–0.76)	1.7
	1.2	16.39	8.04	< 0.001	−18.34	−21.23	NS	34.73	29.27	NS	7.6 (±3.79)	0.60 (0.54–0.65)	2.2
	1.6	17.86	6.13	< 0.001	−18.38	−24.64	< 0.001	36.24	30.77	NS	8.4 (±4.16)	0.56 (0.49–0.61)	2.4

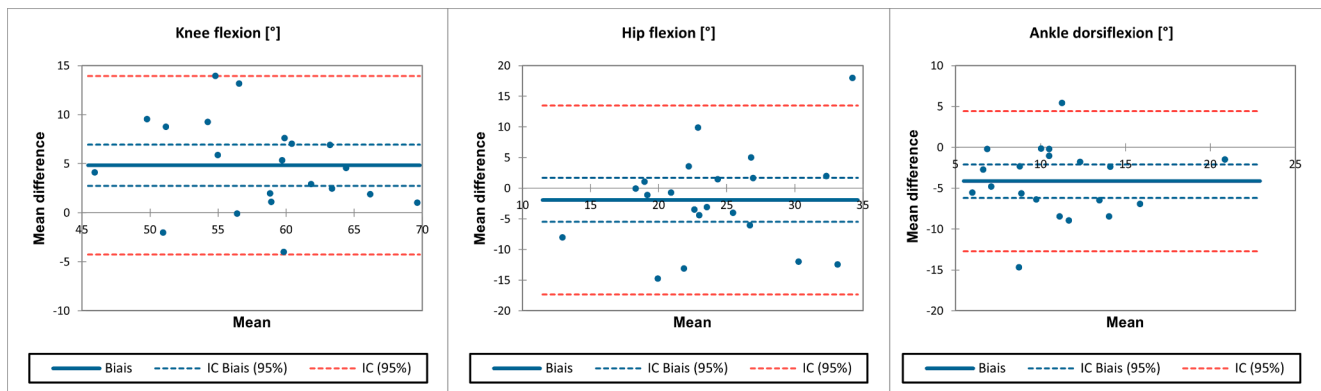


Fig. 2. Bland-Altman plots for validity. X-axis is the average of IMU and MOCAP measurements while y-axis is the difference between the two systems. Solid blue line represents the bias and dotted red lines are the 95 % limits of agreement.

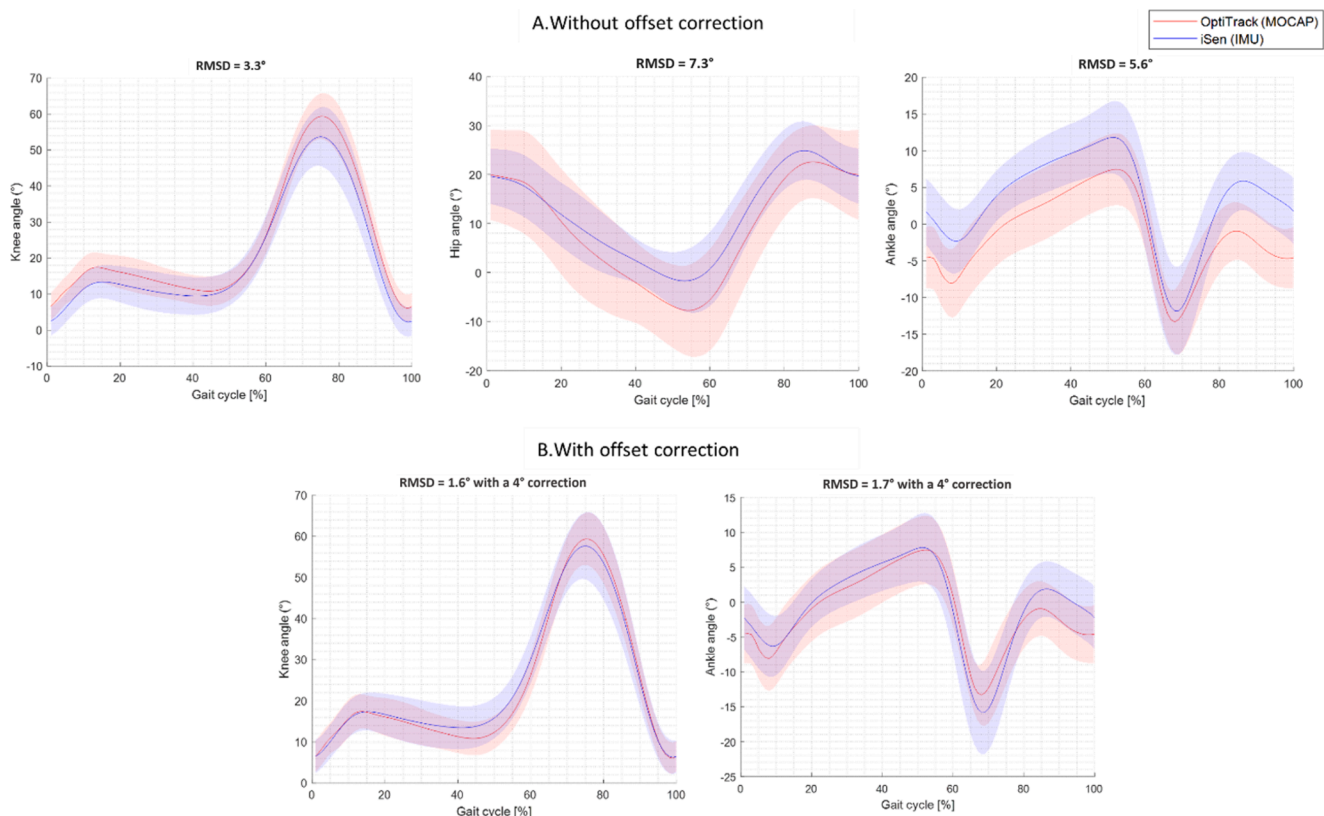


Fig. 3. Joint angle waveforms at 0.8 m/s without offset correction (A) and with offset correction (B). Represented gait cycles are the average gait cycles of the 22 subjects. Red lines are from OptiTrack system and blue lines are from the iSen system. Standard deviation areas are also represented.

Robert-Lachaine et al [18] reported similar values for knee flexion/extension at similar speeds (RMSD = 3.2° at 0.8 m/s and RMSD = 3.4° at 1.2 m/s) with an IMU system focused on 3D knee kinematics calibrated with a double-pose. Max and min biases were under the threshold of clinical significance (error tolerance of 5°, except for the ankle dorsiflexion at 1.2 m/s (bias = -8.4°) and at 1.6 m/s (bias = -11.7°) and hip extension (bias = -7°)). Hip waveform was quite similar to Bessone et al who compared Vicon optoelectronic system and aktos-t IMU [7]. Aktos-t IMU and iSen IMU both overestimated hip extension by 7°. One reason that could explain hip extension tolerable accuracy (RMSD = 7.3°) could be related to the MOCAP biomechanical model. Although MOCAP is considered as “gold standard” in human motion analysis, it can also be influenced by soft tissues artefacts and markers positioning [19]. Soft tissue artefact, due to the relative motion between the bone and the

markers, is the main source of error in the estimation of hip joint center location [20].

The agreement between iSen and OptiTrack, was also interpreted using Lin's concordance correlation coefficient, according to the following criteria ([21,22]): excellent (0.75 – 1.00), good (0.60 – 0.74), fair (0.40 – 0.59), and poor (<0.40). LCC were found to be excellent for all joints (0.93 and 0.81 for knee and hip respectively), regardless of the speed, except for the ankle. Ankle measurement were good at 0.8 m/s (LCC = 0.72), and 1.2 m/s (LCC = 0.60), and fair at 1.6 m/s (LCC = 0.56). Poor ankle measurement has been also reported by Eltoukhy et al [23] (LCC on ankle ROM = 0.01 at 1.3 m/s and -0.01 at 1.6 m/s). In the current study protocol, IMU sensor had been added on the heel assuming a better precision in ankle measurement. However, results did not meet expectations of showing better accuracy than in previous studies. A

Table 2

Bland-Altman biases and limits of agreement (LoA). Minimum (min), maximum (max) and range of motion (ROM) values for knee flexion/extension, hip flexion/extension and ankle dorsiflexion/plantarflexion are presented at each speed.

Joint (Sagittal plane)	Speed [m/s]	Max [°] (LoA)	Min [°] (LoA)	Rom [°] (LoA)
Knee	0.8	4.83 (-4.27; 13.94)	0.46 (-5.41; 6.34)	4.37 (-2.24; 10.97)
	1.2	4.59 (-5.03; 14.21)	-0.1 (-6.30; 6.10)	4.62 (-14.09; 3.17)
	1.6	4.30 (-7.12; 15.73)	1.34 (-9.26; 11.94)	2.97 (-7.61; 13.54)
Hip	0.8	-1.92 (-17.36; 13.51)	-7.11 (-25.52; 11.30)	5.19 (4.93; 15.30)
	1.2	-1.31 (-17.93; 15.31)	-7.26 (-25.884; 11.35)	5.95 (-2.28; 14.19)
	1.6	1.74 (-14.15; 17.63)	-3.10 (-22.04; 15.84)	4.84 (-9.99; 19.67)
Ankle	0.8	-4.15 (-12.73; 4.43)	-0.26 (-12.86; 12.34)	-3.89 (-10.75; 2.97)
	1.2	-8.35 (-15.89; -0.801)	-2.89 (-14.83; 9.06)	-5.46 (-14.09; 3.17)
	1.6	-11.73 (-20.84; -2.61)	-6.26 (-18.92; 6.39)	-5.47 (-13.13; 2.20)

possible explanation could be that the treadmill might have created interferences with IMU magnetometers, with strongest consequences on ankle joint estimations. Indeed, De Vries et al [24] have shown that, when getting closer to the magnetic source, the disturbance effect was stronger (until 5 cm above the source). This could partially explain the lowest accuracy in ankle joint between IMU and MOCAP. Moreover, IMU on the foot were closer to the treadmill motors than other sensors. Indeed, motors are located at the front of the treadmill where participants were asked to walk to increase detection of reflective markers by the cameras. Robert-Lachaine et al [1825] also highlighted magnetic disturbances effect as a source of inaccuracy between Xsens IMU and Optitrack MOCAP.

Accuracy between the two systems was stronger when adjusting waveforms and correcting them with an existent offset. Indeed, due to their microelectromechanical architecture, raw IMU sensor measurements are prone to noise and non-zero biases [6]. Also, offsets and errors might come from ISEN kinematic algorithms and biomechanical model as an integration step of measurements is typically necessary to obtain joint kinematics, resulting in drifting estimates of sensor orientations and joint kinematics [6]. These errors could increase RMSD and reduce accuracy of the system. A $+4^\circ$ correction was then applied on knee gait cycle waveform while ankle waveform was reduced by 4° at 0.8 m/s to compensate the offset. This offset correction increased accuracy as shown by lower RMSD after correction (at 0.8 m/s, RMSD knee = 1.6° ,

RMSD ankle = 1.7°) compared to before correction (at 0.8 m/s, RMSD knee = 3.3° , RMSD ankle = 5.3°). Nüesch et al [26] also applied an offset correction on gait kinematic parameters when subject walked at self-selected speed (1.37 ± 0.13 m/s). When adjusting the offset, data showed clinically acceptable agreement (RMSD knee = 7.6° and 5.0° respectively before and after offset correction, RMSD hip = 9.6° and 3.3° , RMSD ankle = 4.5° and 2.5°). Berner et al [12] results also showed good accuracy when applying offset correction (RMSD ranged between 6.7° – 15.0° before offset correction and RMSD $< 3^\circ$ when applying correction). Consequently, when the offset can be known, it could be applied to the waveforms to increase accuracy or at least to have the real difference between the two systems of interest.

The effect of walking speed on the ISEN accuracy was quantified. Results showed that speed can reduce the accuracy of IMU as shown by the decrease in LCC and the increase in RMSD in high speed conditions. RMSD increase was highlighted by Jeon and Lee [27] (RMSD on knee flexion/extension = 2.70° at 0.75 m/s, 4.29° at 1.5 m/s and RMSD on ankle dorsiflexion/plantarflexion = 2.18° at 0.75 m/s and 5.48° at 1.5 m/s). It was concluded that errors increased in proportion to the magnitudes of the external acceleration. Jeon and Lee also demonstrated that, despite the use of a Kalman filter to compensate the effect of acceleration in fast motion, accuracy of the IMU was still impacted. Bergmann et al [28] also reported the increase of the acceleration magnitude with speed as the principal cause of inaccuracy of the IMUs. Both iSen and OptiTrack systems similarly detected kinematic differences induced by gait speed. Results highlighted IMU's ability to quantify subtle changes in

Table 3

Repeatability of the IMU. ICC (2,1) values for test–retest reliability are presented.

Joint	Speed [m/s]	Max [°] (95% CI)	Min [°] (95% CI)	ROM [°] (95% CI)
Knee Flexion/Extension	0.8	0.95 (0.88–0.98)	0.91 (0.77–0.96)	0.89 (0.72–0.95)
	1.2	0.91 (0.78–0.97)	0.85 (0.62–0.94)	0.89 (0.72–0.96)
	1.6	0.93 (0.82–0.97)	0.88 (0.70–0.95)	0.94 (0.84–0.97)
Hip Flexion/Extension	0.8	0.96 (0.90–0.98)	0.85 (0.62–0.94)	0.87 (0.68–0.95)
	1.2	0.82 (0.54–0.93)	0.80 (0.49–0.92)	0.81 (0.52–0.92)
	1.6	0.88 (0.70–0.95)	0.68 (0.21–0.87)	0.88 (0.71–0.95)
Ankle Dorsiflexion/Plantarflexion	0.8	0.86 (0.64–0.94)	0.85 (0.61–0.94)	0.77 (50.42–0.91)
	1.2	0.78 (0.43–0.92)	0.93 (0.81–0.97)	0.89 (0.71–0.96)
	1.6	0.75 (0.37–0.90)	0.91 (0.78–0.97)	0.84 (0.59–0.94)

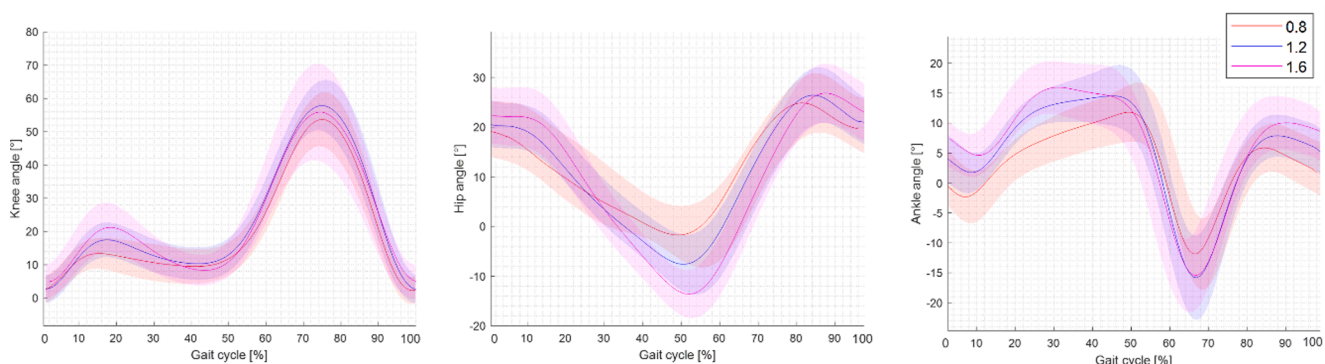


Fig. 4. Joint angle waveform of the IMU system at the three walking speeds. 0.8, 1.2 and 1.6 m/s are respectively represented with red, blue and magenta lines. Standard deviation areas for each speed are also represented.

Table 4

Sensor drift of the IMU at a speed of 1.6 m/s.

Joint (Sagittal plane)	IMU comparison				IMU and MOCAP comparison			
	Max angle difference [°]	Min angle difference [°]	Rom angle difference [°]	RMSD [°]	Max Relative error [°]	Min Relative error [°]	Rom Relative error [°]	RMSD Relative error [°]
Knee	−0.61	−0.64	0.18	1.78	0.33	0.59	−0.41	0.26
Hip	−0.49	−0.78	0.29	1.78	1.11	0.43	0.68	0.59
Ankle	0.04	0.49	−0.45	1.21	−0.14	0.16	−0.30	0.36

kinematic parameters similarly to MOCAP system. Increasing bias with speed on ankle joint angle could be attributed to the high sensibility of IMU gyroscope to vibrations [29,30] induced by foot hitting the treadmill, especially at high speed. Reported ankle dorsiflexion/plantarflexion accuracy was most affected by speed. Moreover, ICC decrease with gait speed highlighted a loss of repeatability, even though ICC remained good (ICC > 0.75 except for hip extension at 1.6 m/s: ICC = 0.68). Hip extension estimation provided by the IMU was less accurate and less repeatable at high speed.

MOCAP markers were placed on precise bony landmarks contrary to IMU sensors, which were positioned more randomly with an easy setup. Indeed, IMU protocol replaced, among others, markers located on the epicondyles with markers on both shanks and thighs. ICC were calculated to assess whether IMU placement influenced the measurements when repeating the same test, two weeks apart. ICC showed good to excellent repeatability (ICC > 0.75 and ICC > 0.90 for knee flexion), according to the guideline of Koo et al [31] for all joints and all speeds except for hip extension at 1.6 m/s (ICC = 0.68) which exhibited moderate repeatability. iSen system demonstrated a good ability to produce consistent results, which is essential for a clinical use. The easy and quick setup of the sensors on the adhesive scratch bands is less time consuming than placing reflective markers with an extreme precision and yet, measures were highly repeatable. These results highlight IMU potential to replace optoelectronic systems in field environment acquisitions.

However, one of the major limitations of using inertial sensors instead of MOCAP is the integration drift induced by the gyroscope. Although participants are remaining still, gyroscopes tend to present a non-zero value induced by the electronic noise of the sensor and creating an offset. Offset errors increase over time and are responsible for kinematic measurements inaccuracies. In the current study, speed of 1.6 m/s was evaluated to report sensor drift with time, as errors showed in increase with walking speed. The evaluation of sensors drift on knee, hip and ankle angles showed excellent results (Table 4). Average gait cycle was identical between first and last 30 s (RMSD < 2°). Relative error on RMSD values highlighted a non-significant increase of kinematics discrete parameters within the three-minute acquisitions. IMU system seems to deliver stable output data over time. Results align with existing gyroscopes presenting drift biases in the order of 0.05°/h [32]. Drift effect can be neglected during a few minutes of continuous use. Regular calibration, before launching each three-minute acquisition, to correct drift offset was well integrated by the iSen software. Drift errors were effectively compensated in IMU algorithms to provide an accurate motion follow-up.

One of the limitations of this study is the focus of inertial sensors measurements in the sagittal plane. Future work should include validation in the three axes. However, analysis of adduction/abduction and rotation angles have demonstrated significant differences with MOCAP systems [33,34]. Additional studies should also focus on spatiotemporal variables [35,36] and on functional movements (squats, jumps, sit-to-stands) [37,38]. Another issue to raise is that the evaluation of IMU performance was done on a treadmill where magnetic field was distorted by the metallic components of the treadmill and the motors. Using inertial sensors in a more ecological environment might provide a better accuracy of the system.

Finally, the iSen system is good in measuring hip, knee and ankle

joints compared to MOCAP system (LCC > 0.6). Even if the accuracy is lower in ankle joint, the measures are still acceptable. Also, the system is repeatable in measuring lower limb kinematics (ICC > 0.6) even if its accuracy can be influenced by walking speed (as shown by the decrease in LCC and the increase in RMSD in high speed conditions).

5. Conclusion

In conclusion, iSen inertial measurement unit provides an acceptable clinical tool for evaluating sagittal plane lower limb kinematic parameters during gait. Comparison of iSen IMU against OptiTrack MOCAP demonstrated good and tolerable agreement respectively at low and high speed. Correction of the offset is recommended to achieve better estimation of knee and ankle joint angles. IMU can be used accurately and repeatably by clinicians to perform gait analysis, accepting a 5° error, at low speed. Current results are encouraging the research and development of inertial systems for rehabilitative applications.

6. Institutional review board statement

The study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Comité Ethique Azuréen (CEA).

7. Informed consent statement

Informed consent was obtained from all subjects involved in the study.

CRedit authorship contribution statement

Elodie Piche: Conceptualization, Methodology, Resources, Writing – review & editing. **Marine Guilbot:** Formal analysis, Investigation, Resources, Data curation, Writing – original draft. **Frédéric Chorin:** Methodology, Validation, Visualization, Supervision. **Olivier Guerin:** Project administration, Funding acquisition. **Raphaël Zory:** Methodology, Validation, Visualization, Supervision. **Pauline Gerus:** Conceptualization, Methodology, Visualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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